# Automated Extraction and Validation of Antenna Radiation-Pattern Phi Cuts from Compact Antenna Test Range Raster Data

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***Abstract*—Compact Antenna Test Ranges (CATR) are widely used for far-field validation of antennas under test (AUT) because they emulate plane-wave illumination in a controlled, reflection-free indoor environment. A standard CATR measurement procedure records the AUT response as a two-dimensional angular raster (a grid over azimuth and elevation, or over $u$–$v$/direction-cosine space) and subsequently extracts one-dimensional angular "cuts" – for example principal-plane cuts at Phi $= 0^\circ, 30^\circ, 90^\circ, 180^\circ$ – for pattern verification against specification masks and prior measurement or simulation data. Directly commanding the positioner to slew along a fixed physical cut is, in practice, compromised by mechanical wobble of the positioning system, which prevents an exact, repeatable boresight-aligned cut, and by the fact that fine angular step sizes (needed for accurate sidelobe and beamwidth characterization) make a dedicated slow cut-by-cut scan prohibitively time consuming. This paper presents a MATLAB-based desktop application, the *Antenna Cut Analysis Suite*, that solves this problem by acquiring only a single, efficient raster scan of the AUT and then computationally reconstructing arbitrary Phi/Azimuth/Elevation cuts from the raster using two-dimensional interpolation. The software further validates the reconstructed cuts against independently recorded reference cut files (in the industry-standard GRASP-style `.asc` format) through automated peak, error, and beamwidth statistics. The paper describes the CATR measurement problem being addressed, the raster (`.grd`) and cut (`.asc`) file formats involved, the parsing and interpolation algorithms implemented in the tool, and its complete functional feature set, and presents a validation methodology comparing raster-derived cuts against measured reference cuts.**

***Index Terms*—Compact Antenna Test Range (CATR), far-field measurement, antenna pattern cuts, raster scan, GRD file, ASC file, bilinear interpolation, boresight alignment, pattern validation, MATLAB**

## I. Introduction

Accurate characterization of an antenna's far-field radiation pattern is a mandatory step in the qualification of any antenna, whether for terrestrial, airborne, or satellite payload use. A **Compact Antenna Test Range (CATR)** achieves this by using a shaped reflector (or a dual-reflector/lens system) to collimate the field radiated by a feed into a quasi-plane wave over a compact quiet zone, allowing far-field conditions to be emulated within an indoor anechoic chamber rather than requiring a physically large outdoor or far-field range. The Antenna Under Test (AUT) is mounted on a precision positioning system inside the quiet zone, and the range receiver records the co-polar and cross-polar response as the positioner steps the AUT (or, equivalently, steps the measurement angle) over the angular region of interest.




Fig. 1: Photograph of the CATR anechoic chamber, reflector, and AUT positioner.

For validation purposes, the AUT pattern is not examined as a full two-dimensional data set but rather along specific one-dimensional slices called **cuts**. The most common convention is a set of **Phi cuts**, where the elevation/theta angle is swept from $-\theta_{max}$ to $+\theta_{max}$ at a fixed azimuthal (Phi) plane, taken at a small number of standard planes such as Phi $= 0^\circ$ (principal E- or H-plane), $30^\circ$, $90^\circ$, and $180^\circ$, in addition to any application-specific planes the antenna designer requires for a given lobe, null, or scan-angle requirement. Each cut is judged against a specification template for parameters such as boresight gain, sidelobe level, cross-polar discrimination (XPD), and $-3$ dB beamwidth.

## II. Problem Statement

While conceptually simple, obtaining these cuts directly by commanding the CATR positioning system presents three practical difficulties that motivate the present work:

1) **Positioner wobble and boresight uncertainty.** Mechanical backlash, bearing tolerance, and gravitational sag in the multi-axis positioner introduce small angular wobble as the AUT is slewed. As a result, a physically commanded "Phi $= 0^\circ$" cut does not reliably pass exactly through the true electrical boresight of the AUT, and repeated cuts at nominally the same Phi angle are not perfectly repeatable.

2) **Slow acquisition at fine angular resolution.** High-resolution cuts (small angular step size), which are required to correctly resolve narrow sidelobes and to accurately locate $-3$ dB beamwidth points, are time consuming to acquire directly, since each cut is a separate, slow, dedicated positioner sweep with settling time at every step.
3) **Inflexibility to arbitrary or late-requested cut angles.** Once the range time is booked for a fixed set of cuts, obtaining an additional cut at an angle that was not originally planned (e.g., a cut through a particular grating lobe or scan beam discovered only during data review) requires additional range time.

The approach adopted here is to instead record a single, denser **raster scan** of the AUT response over a rectangular grid of azimuth and elevation (or $\theta_x/\theta_y$) angles. Because the raster is acquired as one continuous two-dimensional scan rather than as many independent one-axis sweeps, it is significantly faster to record at a given resolution, and any one-dimensional cut through the pattern – including cuts at Phi angles not explicitly commanded during the measurement – can subsequently be reconstructed in post-processing by interpolating the raster along the desired angular line. The reconstructed cuts are then validated against a limited number of independently measured reference cuts (`.asc` files) to confirm that raster-derived and directly measured data agree to within an acceptable error budget.

## III. System Overview

The *Antenna Cut Analysis Suite* is implemented as a self-contained MATLAB App (`antenna_cut_analyzer.m`, MATLAB R2021b+, built on `uifigure`/App Designer-style components) that implements the full raster-to-cut-to-validation workflow described above. At a high level the tool performs four operations:

1) Parses the raster far-field data file exported by the range controller (`.grd` format).
2) Reconstructs arbitrary Azimuth, Elevation, or Radial (Phi) cuts from the raster grid using 2-D interpolation.
3) Parses independently supplied reference/measured cut files (`.asc` format, including multi-frequency, multi-file batch loading) for use as ground truth.
4) Overlays and statistically compares the raster-derived cuts against the reference cuts, and exports the results as figures and re-generated `.asc` files.

The application is organized around two working tabs – *2-D Cut Analysis* and *Cut Comparison* – fed by a left-hand control panel for file loading and sweep configuration, a central plotting workspace supporting 1/2/4-way split views, and a bottom status/log bar.

## IV. Data Formats

### A. Raster Input: the GRD File

The `.grd` file is a plain-text, whitespace/line-oriented far-field raster format. Its structure, as consumed by the parser, is:

- An arbitrary number of leading comment/header lines (beginning with `+` or `#`), skipped until the first purely numeric line.
- A *file-type* code line.
- A parameter line whose first and third integers give, respectively, the number of frequencies `NSET` and the number of field components `NCOMP` (1 for scalar/co-only, 2 for co- and cross-polar complex pairs).
- Per frequency block: a beam-reference coordinate line; a grid-limits line `(XS, YS, XE, YE)` giving the angular extent; a grid-size line `(NX, NY)`; followed by `NX`×`NY` data rows, each containing the real and imaginary parts of component 1 (and, if `NCOMP` $\geq 2$, component 2).

### B. Reference/Validation Input: the ASC File

The `.asc` file is a GRASP-style measurement-format text file that can bundle several `//TEST_FREQ` frequency blocks, each optionally tagged with a `//FILE_POLARISATION` marker (co- or cross-polar) and a `//SCAN_POS` marker that introduces the numeric angle/amplitude data rows for that block. A single `.asc` file may therefore contain several frequencies for one Phi cut, and a full validation set is typically supplied as one file per Phi angle (with the Phi value embedded in the filename, e.g. `...PHI90...`).

## V. Parsing and Analysis Methodology

### A. GRD Parsing Algorithm

The GRD parser (`parse_grd`) reads the file into memory as a cell array of trimmed, non-empty lines and then performs a single forward pass with a line cursor: header/comment lines are skipped based on their leading character, and the file-type and `(NSET, ..., NCOMP)` parameter line are read with `sscanf`. For each of the `NSET` frequency blocks, the grid limits and grid size are read, azimuth and elevation axis vectors are generated with `linspace` over the stated limits, and the following `NX`×`NY` rows are read and reshaped in row-major order into the corresponding azimuth/elevation grid location. For every sample the complex real/imaginary pair for each component is converted to a magnitude in decibels,

$$\mathrm{Mag}_{\mathrm{dB}} = 20\log_{10}\Big(\max\Big(\sqrt{\Re^2+\Im^2},\,10^{-12}\Big)\Big), \qquad (1)$$

the $10^{-12}$ floor being applied purely to avoid a $\log(0)$ singularity for numerically zero samples. Each frequency block is stored as a MATLAB `table` with columns for azimuth, elevation, and the co-/cross-component magnitude and complex parts, indexed as `data_by_freq.f1, f2, ...`.

### B. Grid Reconstruction

Because the raw table is a flat list of (azimuth, elevation, magnitude) samples, `build_grid` first extracts the sorted set of unique azimuth and elevation values present in the scan and then re-assembles a dense 2-D matrix $G[\mathrm{el},\mathrm{az}]$ by matching each unique (azimuth, elevation) pair back to its magnitude sample (tolerance $10^{-9}$ on the coordinate match, with any structurally missing sample defaulted to $-120$ dB). This regularized grid is the basis on which all subsequent cut extraction

and interpolation operate, and is rebuilt independently for the co- and cross-polar component and for every frequency block, since the AUT response, and hence the useful grid resolution, generally differs slightly between the two.

### C. Cut Extraction and Interpolation

Three cut types are supported, selected by the user before generation:

*a) Azimuth cuts (fixed elevation):* The requested elevation is snapped to the nearest available raster row using a minimum-absolute-difference index search (`nearest_idx`), and the corresponding row of the grid is returned directly (no interpolation is needed since the cut lies exactly along a raster axis).

*b) Elevation cuts (fixed azimuth):* Symmetric to the above, operating on grid columns.

*c) Radial / Phi cuts:* This is the general case relevant to arbitrary Phi-plane cuts, since a constant-Phi radial line through the origin of an azimuth-elevation raster does not, in general, lie along either raster axis. The tool parametrizes the desired cut as a straight line

$$x(\theta) = \theta\cos(\phi), \qquad y(\theta) = \theta\sin(\phi), \qquad \theta \in [-\theta_r, \theta_r], \tag{2}$$

sampled at 501 equally spaced points over the requested angular half-range $\theta_r$, and clips the resulting $(x, y)$ query points to stay within the raster's azimuth/elevation extent. A `griddedInterpolant` object is built over the regularized $(\mathrm{el}, \mathrm{az})$ grid using *bilinear* interpolation (MATLAB `'linear'` method) with extrapolation disabled, and the interpolant is evaluated at every $(x(\theta), y(\theta))$ pair to obtain the interpolated co-/cross-polar level along the Phi line. Any query point that still falls outside the convex hull of the raster (returned as `NaN` by the interpolant) is clamped to the raster's global minimum level rather than left undefined, which keeps the resulting curve continuous for plotting and error computation. This bilinear-interpolation step is the computational core of the tool: it is precisely what allows a cut at *any* Phi angle – not only the angles physically commanded during acquisition – to be reconstructed from a single raster pass, directly addressing the "arbitrary angle" and "fine step size" limitations described in Section II.

For all three cut types, the extracted level is normalized to the frequency block's own raster peak $P_k = \max(G)$ before storage, i.e. every cut is initially expressed in dB relative to the peak of the full raster:

$$\mathrm{Level}_{\mathrm{dB}} = G_{\mathrm{cut}} - P_k. \tag{3}$$

### D. Angle List Generation

For Azimuth/Elevation cuts, the set of cut angles is generated outward from $0^\circ$ in both directions at the user-specified step size and clipped to the raster's own angular extent (`stepFrom0`), guaranteeing that only angles actually covered by the measured raster are ever requested. For Radial/Phi cuts, the angle list is simply $0^\circ, \mathrm{step}, 2\cdot\mathrm{step}, \ldots, (360-\mathrm{step})^\circ$, so that a full azimuthal sweep of principal and intermediate Phi cuts – including the $0^\circ/30^\circ/90^\circ/180^\circ$ planes used for validation in Section VIII – is produced automatically for any requested angular resolution, all from the same single raster acquisition, with no additional range time.

### E. Boresight / Peak-Offset Correction

To compensate for the positioner-wobble-induced boresight uncertainty discussed in Section II, an optional "Force Offset Peak Alignment" mode (`applyPeakCorrection`) can be toggled without re-generating the cuts: it locates the peak sample of a given cut, then rigidly shifts the angle axis so the peak sits at $0^\circ$ and shifts the level axis so the peak reads exactly 0 dB,

$$\theta' = \theta - \theta_{\mathrm{peak}}, \qquad L' = L - L_{\mathrm{peak}}. \tag{4}$$

This provides a self-consistent, mechanically-independent boresight reference for every cut, decoupled from any physical mounting or positioner zero-offset error.

### F. $-3$ dB Beamwidth Extraction

The $-3$ dB points on either side of the cut's peak are found (`find3dBPoint`) by locating the last sample *left* of the peak and first sample *right* of the peak whose level crosses $P_{\mathrm{peak}} - 3$ dB, and then linearly interpolating between the two bracketing raw samples to obtain the sub-sample-resolution crossing angle, i.e. for bracketing samples $(x_1, y_1), (x_2, y_2)$:

$$\theta_{-3\mathrm{dB}} = x_1 + (P_{\mathrm{peak}} - 3 - y_1)\,\frac{x_2 - x_1}{y_2 - y_1}. \tag{5}$$

### G. ASC (Reference) Parsing

The reference-cut parser (`parseASCMultiBlock`) implements a small line-oriented state machine: it scans the file for `//TEST_FREQ` tags (which close out any block currently being accumulated and open a new one, with the frequency value extracted by regular expression), `//FILE_POLARISATION` tags (which set the current block's polarity to co- or cross-polar by keyword match), and `//SCAN_POS` tags (which mark the start of the numeric angle/amplitude rows that follow). Every subsequent non-comment numeric line is accumulated into the current block's data buffer until the next tag or end of file, after which the block's angle/level arrays are sorted by angle and stored. If no recognizable tag structure is found at all, the parser falls back to a simple two-column numeric read (`parseASCDataPlain`) so that plain, untagged cut files can still be loaded. The Phi angle associated with a given `.asc` file is auto-detected from the filename using a small cascade of regular-expression patterns (`PHI`*n*, *n*`deg`, `_`*n*`_`, `@`*n*), and files for which no angle can be detected are auto-numbered and reported once in a single batch summary rather than blocking a bulk load of 20+ files with per-file dialogs – an explicit optimization for loading a full validation set in one action.

### H. Cut Comparison and Error Statistics

Once a raster-derived cut and its corresponding reference `.asc` cut are both available at the same Phi angle (matched via `findClosestAngle` with a 2° tolerance) and at the same selected frequency, `calcErrors` resamples both curves onto a common, overlapping angular domain (501 points, linear interpolation with extrapolation to handle slightly differing endpoints) and reports:

- **Maximum absolute error** $\max|L_{\mathrm{GRD}} - L_{\mathrm{ASC}}|$ over the overlap region,
- **Mean absolute error** $\mathrm{mean}|L_{\mathrm{GRD}} - L_{\mathrm{ASC}}|$,
- **Peak angular shift**, the difference between the two curves' peak locations.

These three figures of merit constitute the quantitative validation metric used to confirm that a raster-reconstructed Phi cut reproduces an independently, directly measured cut to within acceptable tolerance.

## VI. Software Functionality

Beyond the parsing/interpolation/validation core described above, the implemented application provides the following functional capabilities, organized by area:

### A. File I/O and Batch Handling

- Loading of a single `.grd` raster file, with automatic detection of the number of frequency blocks (NSET) and field components (NCOMP).
- Bulk, multi-select loading of 20+ `.asc` reference files in one dialog, with automatic per-file Phi-angle detection from filename and a single consolidated summary for any files where the angle could not be auto-detected (rather than a blocking per-file prompt).
- Re-export of every generated raster cut back out as a GRASP-style `.asc` file (`writeASCBlock`/`exportAllCutsToASC`), bundling all NSET frequencies for each Phi cut into a single file, mirroring the format of the original measured reference files.

### B. Sweep Configuration and Cut Generation

- Selectable cut type: Azimuth cuts (vary elevation), Elevation cuts (vary azimuth), or Radial/Phi cuts.
- User-configurable angular step size and angular half-range.
- Selectable field component (Comp1/Comp2, i.e. co-/cross-polar) and, independently, selectable frequency block for both raster data and comparison.
- "Add Custom Angle" to generate one additional on-demand cut, across all frequencies, after the main sweep has already run.

### C. Visualization and Workspace

- 1/2/4-way split-screen plotting workspace with optional synchronized zoom across panels.
- Overlay mode to superimpose multiple generated cuts on one axes for direct visual comparison, e.g. a full azimuthal sweep of Phi cuts generated from 0° to 360° in 10° steps and overlaid together.
- Draggable on-plot cursor marker reporting Co-Pol level, Cross-Pol level, and the live Co−Cx (XPD) difference at the cursor position, labeled by trace.
- Automatic peak-gain, −3 dB beamwidth marker (with the power value printed at each −3 dB point), and boresight markers on every cut plot.
- Double-click-editable graph titles and axis labels.
- Light-on-dark "Mission Control" themed control chrome with a publication-style white-background plot area suitable for direct use in reports.

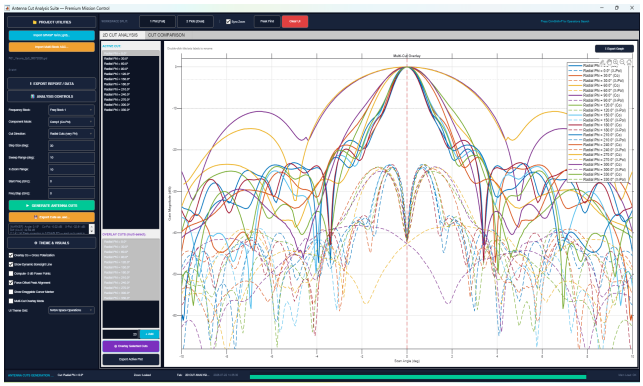


Fig. 2: Overlay of raster-reconstructed Phi cuts swept from $\Phi = 0°$ to 360° in 10° steps, plotted together on one axes, showing Co-Pol and Cross-Pol traces, peak, and −3 dB markers.

### D. Validation / Comparison Tab

- Side-by-side raster-vs-reference plots at each available Phi angle, with a frequency selector that lets the user choose which of several `//TEST_FREQ` blocks bundled inside an `.asc` file the comparison is run against.
- Automatic tabulation of maximum error, mean error, and peak angular shift per compared cut.

## VII. Results and Validation

This section presents the validation of raster-reconstructed cuts against independently measured reference cuts. *(Populate with project-specific data, screenshots, and statistics before submission.)*

### A. Overlay of Reconstructed Cuts

### B. Individual Cuts and Comparison with Measured ASC Data

Figures 3–5 show the raster-derived cut individually at each of three representative validation planes, Phi = 0°, 30°, and 180°. Fig. 6 then overlays the Phi = 0° and 30° raster-derived cuts against the independently measured reference (`.asc`) cuts at the same planes and frequency, for direct comparison.

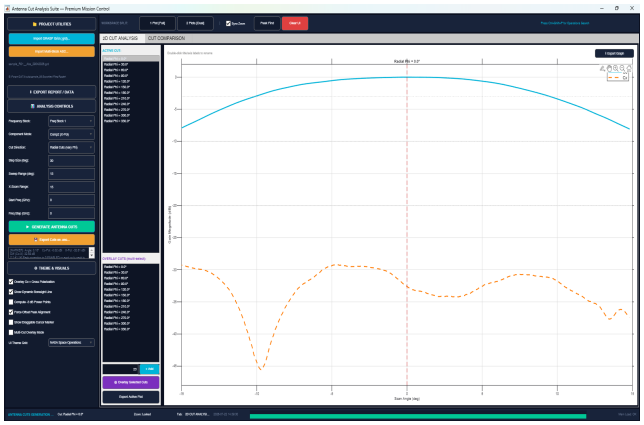


Fig. 3: Phi = 0° cut: raster-derived (Co/Cx).

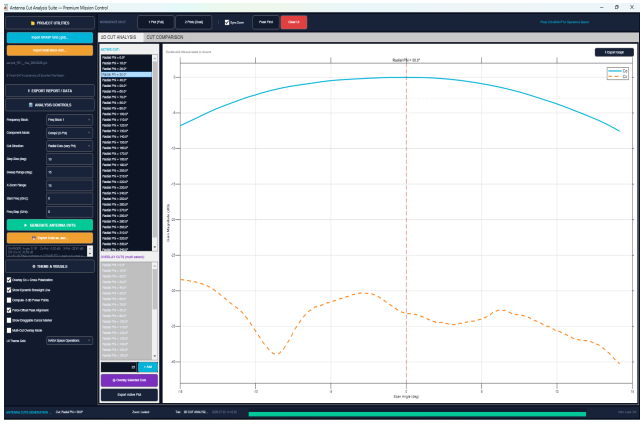


Fig. 4: Phi = 30° cut: raster-derived (Co/Cx).

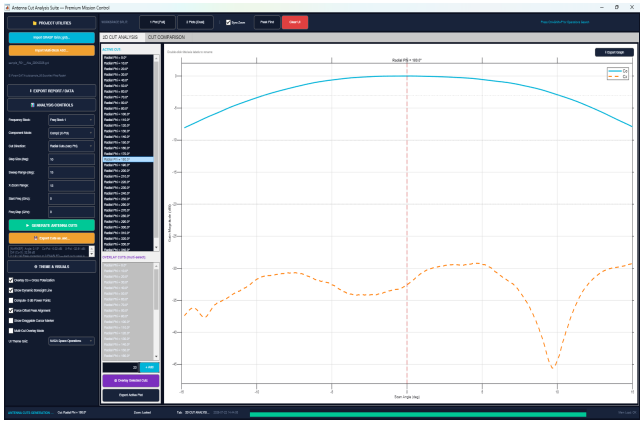


Fig. 5: Phi = 180° cut: raster-derived (Co/Cx).

## VIII. Discussion

The reconstruction approach shifts the burden of resolving arbitrary-angle, fine-resolution cuts from the CATR positioning hardware onto post-processing interpolation of a single raster acquisition. This directly mitigates the three problems identified in Section II: (i) boresight uncertainty from positioner wobble is addressed independently of the raw mechanical pointing by the peak/offset self-alignment step, rather than

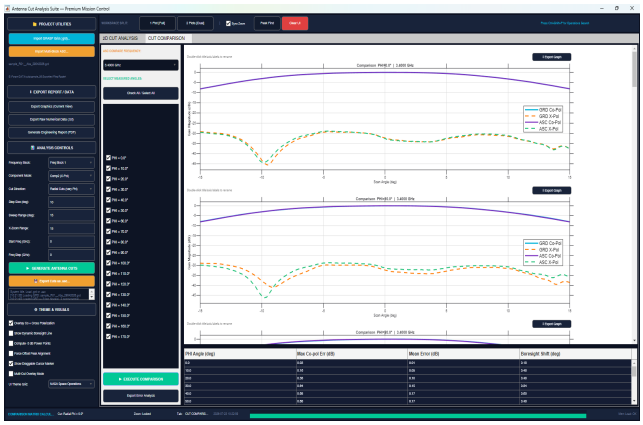


Fig. 6: Comparison of Phi = 0° and 10° raster-derived cuts with measured ASC reference cuts.

requiring a perfectly wobble-free physical cut; (ii) acquisition time no longer scales with the number and resolution of the required cuts, since a single raster produces an arbitrary number of cuts at arbitrary resolution in post-processing; and (iii) any previously unplanned Phi angle can be produced after the fact from the same raster, with no additional range booking. The bilinear-interpolation error inherent to reconstructing an off-axis Phi cut from an on-axis azimuth/elevation raster is quantified directly by the maximum/mean error and peak-shift statistics against the independently measured `.asc` reference cuts, giving an empirical bound on the accuracy of the reconstruction for a given raster grid resolution.

## IX. Conclusion

This paper presented the design, data formats, parsing/interpolation algorithms, and complete functional feature set of a MATLAB-based Antenna Cut Analysis Suite that reconstructs arbitrary far-field Phi/Azimuth/Elevation cuts from a single CATR raster scan and validates the reconstructed cuts against independently measured reference (`.asc`) cuts. By replacing multiple slow, wobble-sensitive, fixed-angle positioner sweeps with one raster acquisition followed by bilinear-interpolation-based cut reconstruction, the tool reduces range time, removes the constraint of pre-selecting cut angles before the measurement session, and provides a quantitative, automated error/beamwidth/peak-shift validation report against measured ground truth. Future work includes extending the interpolation core to higher-order (bicubic/spline) schemes for improved sidelobe fidelity, and adding native $u$-$v$/direction-cosine grid support alongside the current azimuth-elevation grid.


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